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National University of Computer and Emerging Sciences

Lahore Campus

Embedded Systems (EE3006)

Sessional-I Exam

Date: 23rd February, 2026

Course Instructor(s)

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Total Time (Hrs):	1
Total Marks:	30
Total Questions:	2

Roll No ____________ Section ____________ Student Signature ____________

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Instructions:

1. **Do not forget to write your Name and Roll Numbers.**
2. **Make assumptions if you think some data is missing.**

CLO # 1: Explain the fundamentals of Embedded C programming, and practical applications in developing efficient and reliable embedded systems (PLO 2, C1)

Q1: **[4+2+4 marks]**

a. Write a C program that **demonstrates** bit masking techniques to monitor bit 5 of an 8-bit variable var1.

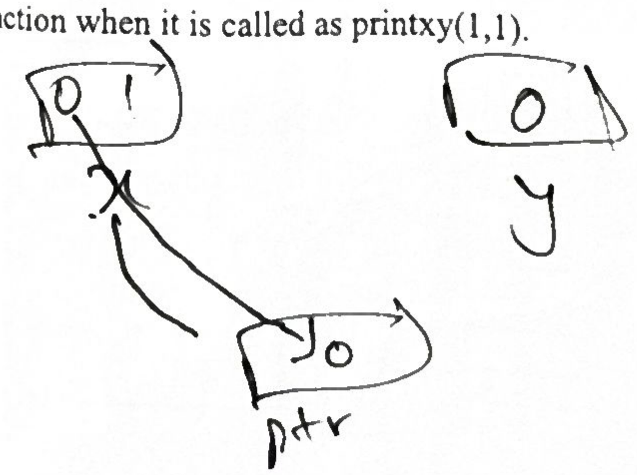
If bit 5 of var1 is HIGH (1), toggle bit 3 of another variable var2.

If bit 5 of var1 is LOW (0), set bit 4 of var2.

Use appropriate bit-wise operators to perform the required operations.

b. **Determine** the output of the following C function when it is called as printxy(1,1).

```
void printxy (int x, int y)
{
   int *ptr;
   x = 0;
   ptr = &x;
   y = *ptr;
   *ptr = 1;
   printf("%d,%d", x, y);
}
```



c. In an embedded system, a memory-mapped register is located at address 0x40004000. The register stores an 8-bit sensor value that may change due to hardware.

i. **Prepare** a macro to define and access this register using appropriate type qualifiers.
ii. **Explain and discuss** why macros are preferred in embedded systems.

CLO # 2: Analyze the architecture of the ARM Cortex-M4 micro-controller, and system-level features relevant to embedded system design (PLO 2, C4)

Q2:

[20 marks]

Configure the TM4C micro-controller with a tri-color LED (Green, Blue, Red). The clock frequency is 16 MHz. Use the bit-specific addressing method in the program.

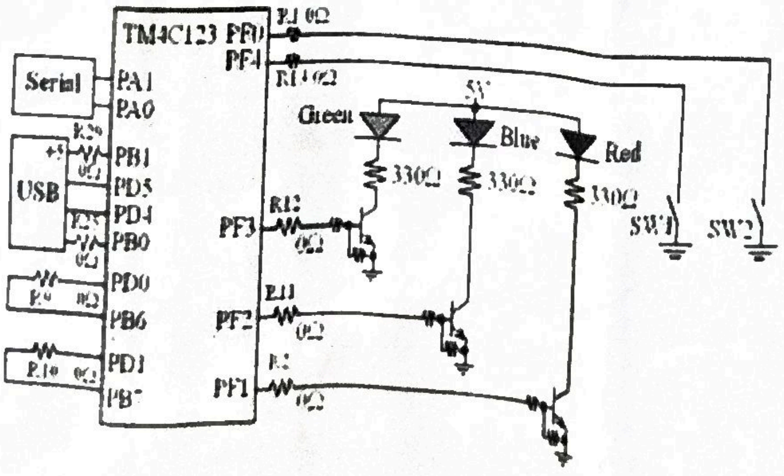
Write a C program to perform the following tasks:

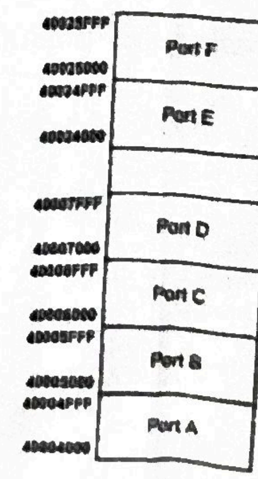
Two switches are connected: SW1 (PF4) and SW2 (PF0).

a) When SW1 (PF4) is pressed, **examine** how the Green and Blue LEDs can be turned ON simultaneously for 2 seconds.
b) When SW2 (PF0) is pressed, **analyze** and **investigate** how the Red LED can be toggled every 5ms.
c) When neither SW1 nor SW2 is pressed, all LEDs should remain OFF.

NOTE:

- Use the SysTick timer to generate accurate delays.
- Ensure that PF0 is unlocked before configuration.
- Use the bit-specific addressing.
- Lock Register unlock value: 0x4C4F434B





Run Mode Clock Gating Control GPIO Register: Base address is 0x400FE000 and offset is 0x608

0104

If we wish to access bit	*Constant*
7	0x0200
6	0x0100
5	0x0080
4	0x0040
3	0x0020
2	0x0010
1	0x0008
0	0x0004

Offset	Name
0x0000	GPIODATA
0x0400	GPIODIR
0x0510	GPIOPUR
0x051C	GPIODEN

Offset	Name
0x520	GPIOLOCK
0x524	GPIOCR